EARLY-PHASE CLINICAL PHARMACOLOGY · BIOSTATISTICS DELIVERABLES PIPELINE · 136 DELIVERABLES

Protocol review → CSR draft

S1 Study Design & Protocol

- Protocol biostat input
- Estimands → endpoints
- Sample-size justification
- Randomization design
- DSMB / SRC decision rules

GATE **Protocol approved**

S2 Planning & Specifications

- Statistical Analysis Plan
- TLF shells / mock-ups
- ADaM & SDTM specs
- Randomization schedule
- Data-review & QC plans

GATE **SAP & specs final**

S3 Data & CDISC

- SDTM datasets + aCRF
- ADaM datasets
- Define-XML + cSDRG/ADRG
- Conformance (Pinnacle 21)
- Database lock + audit trail

GATE **Database lock**

S4 Analysis & QC

- PK conc & NCA TLFs (Phoenix)
- Safety / PD TLFs
- popPK / exposure-response
- Independent double-programming
- Unblinding (controlled)

GATE **Outputs QC-passed**

S5 Reporting & CSR

- CSR methods & results text
- In-text tables / figures
- CSR appendices (16.x)
- Define + guides → eCTD m5
- Cross-doc consistency QC

GATE **CSR draft delivered**

CDISC data-standards track: SDTM → ADaM → Define-XML + cSDRG / ADRG + conformance (runs through Stages 3–5)

Hybrid-AI governance track: data-classification routing · model-freeze records · engine-of-record register · zero-egress attestation (all stages)

🟢 **GREEN · 48** drafts, scaffolds, docs (de-identified)

🟡 **AMBER · 54** dataset / TLF code: AI as one side of double-programming

🔴 **RED · 34** validated engine produces the number; LLM out of the path

3 hard rules: PHI / unblinded never leaves (local frozen model) · every reported number from a validated engine, never an LLM · de-identification unlocks the cloud path (contracted cloud model only).

136 deliverables across 5 stages · full register with every field = the .xlsx · stage-by-stage detail = the Word/PDF reference · standards current mid-2026 (CDISC SDTMIG v3.4 / ADaMIG v1.3 / Define-XML v2.1; ICH E3·E6(R3)·E9(R1)·M11; 21 CFR Part 11).